Converted from Word document

1. Permanent Magnet Moving Coil (PMMC) Instrument

Permanent Magnet Moving Coil (PMMC) Instrument

Construction:

Permanent Magnet → Provides a strong uniform magnetic field

Moving Coil → Placed between magnet poles, wound on aluminum frame

Soft Iron Core → Improves magnetic field strength

Spring (Control Spring) → Provides controlling torque

Pointer & Scale → Shows reading

Damping → Eddy current damping (due to aluminum frame)

Principle:

Works on Lorentz force principle:
A current-carrying conductor placed in a magnetic field experiences a force.

Operation:

When DC current flows through the coil → magnetic field is produced.

Interaction with permanent magnetic field → force acts on coil.

Coil rotates → pointer moves on scale.

Spring provides opposing torque → system stabilizes.

Important Result:

 Deflection is directly proportional to current
→ Uniform scale (very important point for exams)

Advantages:

High accuracy

Uniform scale

Low power consumption

Good damping

Disadvantages:

Works only with DC (Direct Current)

Cannot measure AC directly

Applications:

DC ammeters

DC voltmeters

Galvanometers

4. Moving Iron Instruments (Attraction & Repulsion)

Moving Iron (MI) instruments are commonly used for measuring AC and DC currents and voltages. They work on the principle that a piece of iron gets magnetized when placed in a magnetic field and experiences a mechanical force. MI instruments are of two types:

Repulsion type and Attraction type, each with slightly different operation. Here’s a simple explanation:

1. Repulsion Type Moving Iron Instruments

Construction & Principle:

Contains two iron pieces, one fixed and one movable.

When current flows through the coil, both iron pieces are magnetized with like poles facing each other.

Like poles repel each other.

Function:

The movable iron piece rotates due to magnetic repulsion.

Rotation is opposed by a spring, which gives a deflection proportional to current/voltage.

Used to measure AC or DC.

Characteristics:

Better accuracy at low currents.

Less friction because movable part is light.

2. Attraction Type Moving Iron Instruments

Construction & Principle:

Contains one fixed coil and a movable iron vane/piece.

The coil is energized by the measured current.

The iron is attracted towards the coil due to magnetization.

Function:

The movable iron moves toward the coil.

A spring provides a counter torque, so the deflection is proportional to current/voltage.

Can also measure AC or DC.

Characteristics:

Simpler in construction.

Generally used for low-cost instruments.

Key Difference Between Repulsion & Attraction MI Instruments

Feature

Repulsion Type

Attraction Type

Force

Repulsive (like poles repel)

Attractive (iron pulled to coil)

Accuracy

Higher at low currents

Lower accuracy

Construction

Two iron pieces

One iron piece

Common Use

Precision instruments

Low-cost instruments

5. Shaded Pole Induction Instruments & Energy Meter

1. Shaded Pole Induction Type Instruments

These instruments are mainly used to measure AC current and voltage. They are AC-only instruments.

Construction:

Stator (Magnetic Core): Made of laminated soft iron to reduce eddy current losses.

Coil: A main coil wound on the stator to produce a magnetic field when AC flows.

Shaded Pole: A small copper ring (shading coil) around part of the pole. This creates a delayed magnetic field in that section.

Moving Element (Rotor/Disc): Usually a lightweight aluminum disc mounted on a spindle.

Control Spring: Provides torque opposite to the rotation and brings the disc back to zero.

Dial & Pointer: Shows the measurement reading.

Operation:

When AC passes through the coil, it produces an alternating magnetic field in the core.

The shading coil delays the flux in that portion of the pole. This creates a rotating magnetic field.

The aluminum disc experiences eddy currents due to this rotating flux.

These eddy currents produce torque, causing the disc to rotate.

The rotation is proportional to AC current or voltage. The pointer shows the reading on the dial.

Key Points:

Works only with AC.

Used in AC ammeters, voltmeters, and wattmeters.

High frequency response is limited.

2. Energy Meter (Induction Type)

Energy meters measure electric energy consumption (kWh) in homes, offices, or industries.

Construction:

Stator: Two coils

Current coil (series): Connected in series with load, carries current.

Voltage coil (shunt/parallel): Connected across supply voltage.

Rotor (Aluminum Disc): Mounted on a spindle, free to rotate.

Shading/Aluminum disc: For generating torque via eddy currents.

Brake Mechanism: Usually an aluminum disc eddy current brake with a permanent magnet to control speed.

Register/Counter: Records the number of revolutions → corresponds to energy consumed.

Operation:

When AC flows through the current and voltage coils, a rotating magnetic field is produced.

Eddy currents are induced in the aluminum disc.

These eddy currents generate torque, causing the disc to rotate.

The speed of rotation is proportional to power ($V \times I \times \cos \varphi$).

A braking magnet applies a retarding torque so that the disc rotates at a rate proportional to energy consumed, not instantaneous power.

The counter records energy in kWh.

Key Points:

Works only with AC.

Measures energy, not instantaneous power.

Used in residential and industrial meters.

Difference Between Shaded Pole Instrument & Energy Meter

Feature

Shaded Pole Instrument

Energy Meter (Induction Type)

Purpose

Measures AC current/voltage

Measures energy consumption (kWh)

Moving Part

Aluminum disc

Aluminum disc

Torque Generation

Eddy currents from AC & shaded pole

Eddy currents from current & voltage coils

Application

AC ammeters, voltmeters

Household/Industrial energy measurement

Control Mechanism

Spring

Braking magnet for speed control

□□ 6. Electrical Temperature Sensors

1. RTD (Resistance Temperature Detector)

Construction:

Made of pure metal wire (usually platinum, sometimes nickel or copper) wound on a ceramic or glass core.

Housed in a protective sheath to prevent damage.

Common types: Pt100 (100 Ω at 0°C), Pt1000 (1000 Ω at 0°C).

Principle:

Based on the change in resistance of metal with temperature.

Resistance increases with temperature (positive temperature coefficient).

Operation:

A constant current is passed through the RTD.

The voltage across the RTD changes with resistance.

This voltage is measured and converted into temperature using calibration tables or formulas.

Advantages:

High accuracy.

Stable over long periods.

Good for industrial applications.

2. Thermocouple

Construction:

Made of two different metals joined at one end (measuring/junction).

Common types: Type K (Nickel-Chromium / Nickel-Alumel), Type J (Iron / Constantan).

Principle:

Based on the Seebeck effect: when two different metals are joined, a voltage is generated that depends on temperature difference.

Operation:

Measuring junction is placed at the temperature to measure.

The other end (reference junction) is kept at a known temperature.

The voltage generated is proportional to the temperature difference.

A meter or electronic circuit converts this voltage into temperature reading.

Advantages:

Can measure very high temperatures.

Fast response time.

Simple and robust.

3. Thermistor

Construction:

Made of semiconductor material (metal oxide ceramic).

Shaped as beads, discs, rods, or chips.

Principle:

Based on the change of resistance with temperature.

Can have:

NTC (Negative Temperature Coefficient) → resistance decreases with temperature.

PTC (Positive Temperature Coefficient) → resistance increases with temperature.

Operation:

Connected in a circuit (usually a voltage divider).

The change in resistance alters the voltage across the thermistor.

This voltage is measured and converted to temperature.

Advantages:

Very sensitive to small temperature changes.

Compact and inexpensive.

Good for precision temperature control.

Comparison Table

Sensor Type

Principle

Temp. Range

Accuracy

Response

Application

RTD

Resistance changes w/ T

-200°C to 850°C

High

Medium

Industrial, Lab

Thermocouple

Seebeck effect (voltage)

-200°C to 2500°C

Medium

Fast

High-temp, furnaces

Thermistor

Resistance changes w/ T

-50°C to 150°C

Very High

Very Fast

Electronics, HVAC

7. Electrical Pressure Transducers

Strain Gauge Pressure Transducer

Construction:

A strain gauge (resistive sensor) is attached to a diaphragm.

The diaphragm deforms under applied pressure.

Connected in a Wheatstone bridge circuit to detect resistance changes.

Principle:

Based on strain-resistance relationship:

When the diaphragm is stressed by pressure, the strain gauge stretches or compresses.

Resistance of the strain gauge changes proportionally.

Operation:

Pressure applies force on the diaphragm.

Diaphragm bends → strain gauge deforms.

Resistance change in the gauge produces a voltage change in the Wheatstone bridge.

Output voltage is calibrated to show pressure.

Applications:

Industrial process control, hydraulic systems, aerospace.

2. Potentiometric Pressure Transducer

Construction:

Uses a diaphragm connected to a slider of a potentiometer.

Potentiometer may be linear or rotary type.

Principle:

Pressure deflects the diaphragm → moves the slider → changes potentiometer resistance.

Operation:

Pressure applied → diaphragm moves.

Slider moves along the potentiometer track.

Resistance changes → voltage output changes.

Voltage is calibrated to indicate pressure.

Applications:

Low to medium pressure measurements, laboratory instrumentation.

3. Bourdon Tube Pressure Transducer

Construction:

A curved, hollow tube (C or spiral shape).

One end fixed, the other connected to a mechanical linkage and pointer or electrical transducer.

Principle:

Pressure inside the tube causes tube deformation (straightening or twisting).

Mechanical movement is converted to pointer deflection or electrical signal.

Operation:

Fluid pressure enters the Bourdon tube.

Tube tends to straighten → moves a lever.

Lever moves pointer or electrical sensor (like potentiometer or LVDT).

Reading corresponds to pressure.

Applications:

Analog pressure gauges, hydraulic systems, steam boilers.

4. LVDT (Linear Variable Differential Transformer) Pressure Transducer

Construction:

Core connected to a diaphragm.

LVDT consists of primary coil and two secondary coils.

Principle:

Diaphragm deflection moves the LVDT core.

This causes a differential voltage in the secondary coils → proportional to displacement → proportional to pressure.

Operation:

Pressure applies force on diaphragm.

Diaphragm moves → moves LVDT core.

LVDT generates an AC voltage proportional to core displacement.

Voltage output is calibrated to indicate pressure.

Applications:

High accuracy pressure measurement, aerospace, industrial process control.

Comparison Table

Type

Principle

Accuracy

Applications

Strain Gauge

Resistance changes with strain

High

Industrial sensors, hydraulics

Potentiometric

Diaphragm moves potentiometer

Medium

Lab instruments, low-medium pressure

Bourdon Tube

Tube deformation

Medium

Analog gauges, boilers

LVDT-based

Displacement → voltage

Very High

Precise measurement, aerospace

8. Differential Flow Meters

➤ Differential Flow Meters

Purpose:

Measure flow rate of fluid (liquid or gas) in a pipe.

Work on the principle of pressure difference created by a flow restriction.

Principle:

When fluid flows through a constriction (like orifice or venturi), velocity increases and pressure decreases.

According to Bernoulli’s principle, the pressure difference is proportional to flow rate.

1. Orifice Flow Meter

Construction:

A thin plate with a sharp-edged hole (orifice) placed inside the pipe.

Connected to differential pressure measuring device (manometer, transmitter).

Operation:

Fluid flows toward the orifice → velocity increases, pressure drops at the orifice.

Differential pressure is measured before and at the orifice.

Flow rate is calculated using Bernoulli’s equation and flow coefficient.

Characteristics:

Simple and low cost.

Creates high pressure loss.

Suitable for low to medium accuracy requirements.

2. Venturi Tube Flow Meter

Construction:

A gradually converging section → throat → gradually diverging section in a pipe.

Differential pressure measured between inlet and throat.

Operation:

Fluid enters converging section → velocity increases, pressure decreases.

At throat, maximum velocity → minimum pressure.

Pressure difference between inlet and throat is measured.

Flow rate calculated using Bernoulli’s equation.

Characteristics:

Smooth converging/diverging reduces energy loss.

More accurate than orifice meters.

More expensive and bulky.

Comparison Table

Feature

Orifice Meter

Venturi Tube Meter

Construction

Thin plate with hole

Converging-diverging tube

Pressure Loss

High

Low

Accuracy

Medium

High

Cost

Low

High

Applications

General industrial flow

High accuracy flow measurement, large pipelines